



Official Problem Statement

CareerPilot

Your Agentic Career Co-pilot

Build an AI platform that knows you — hunts jobs, scores your fit, drafts your applications, and builds your learning roadmap.

1. The Problem

Today's job seekers face a fragmented mess: scattered job boards, generic AI tools with no memory of who they are, zero visibility into skill gaps, and no accountability for applications or learning goals.

Your challenge is to build CareerPilot — an end-to-end agentic career co-pilot that solves all of this in one platform. The AI doesn't just answer questions. It actively works for the user, grounded in their real profile.

The Core Idea

A RAG (Retrieval-Augmented Generation) layer over the user's own CV is the single source of truth. Every agent, every recommendation, every cover letter is grounded in this specific user's actual experience — not a hallucinated generic profile. No agent hallucinates the user's background.

2. What You Must Build — The Four Pillars

Pillar 1 — Job Hunter Agent

The agent actively hunts, filters, and presents job opportunities structured for immediate decision-making.

- **Input:** Natural language — "Find me ML internships in Dhaka open this month"
- **Output:** Structured job cards — role, company, salary range, application deadline, location, fit score
- **Reasoning:** Agent explains WHY each result matches (or doesn't) the user's profile, grounded in their CV

Pillar 2 — Profile & Resume Intelligence (RAG Core)

The user's CV is the foundation of everything. It is semantically indexed and queried by every agent.

- User uploads a PDF/DOCX CV or builds one directly inside the platform
- CV is chunked by section: experience, education, skills, projects
- Chunks are embedded and stored in a vector database
- All downstream features — job matching, cover letters, gap analysis — RAG this store

Pillar 3 — Personal AI Assistant

A conversational interface with full context of the user. The assistant knows who they are before they say a word.

Your assistant must handle all of these:

- "Am I ready for this data engineer role?" → Verdict with reasoning, grounded in the user's CV and the JD
- "What skills am I missing for a Google internship?" → Skill gap analysis vs benchmark profiles
- "Build me a 3-month roadmap to become job-ready" → Structured weekly plan with learning resources
- "Draft a cover letter for this job posting" → Personalized letter that references the user's actual experience

Pillar 4 — Productivity & Progress Tracker

Accountability infrastructure that makes the platform useful day-to-day, not just a one-time query tool.

- **Calendar & To-Do:** Calendar view with deadline reminders; to-do items per day/week linked to career goals
- **Goal Setting:** "Apply to 5 jobs this week", "Finish DSA course by Friday", "Update CV by Sunday"
- **Application Tracker:** Kanban board — Applied, Interviewing, Offer, Rejected — with full history
- **Progress Dashboard:** Weekly stats: applications sent, skills added, roadmap % complete, streak counter
- **AI Nudges:** Agent proactively reminds: "You haven't applied this week. Here are 3 openings matching your profile."

3. Required Features

Every submission must include all of the following. Missing items directly affect your score.

Core Technical Requirements

- **RAG architecture grounded in the user's actual CV**
- At least one agent must use external tool calls — web search, a job board API, or scraping
- The AI assistant must demonstrate conversational memory within a session
- Fit scores must be computed programmatically (not just stated)
- The tracker module must include a working calendar and to-do component

Feature Checklist

1. Working web or mobile app with all four pillars implemented or prototyped
2. CV upload and processing pipeline: PDF/DOCX ingestion → chunking → embedding → vector DB
3. Job Hunter Agent with at least one live search and structured card output
4. Fit score: given a job posting, compute a % match against the user's CV with explanation
5. AI Assistant chat with RAG-grounded responses across all benchmark query types
6. Calendar + to-do module with deadline tracking linked to goals
7. Kanban application tracker: Applied / Interviewing / Offer / Rejected
8. Progress dashboard with real data: applications, skills, roadmap progress

4. Deliverables

4.1 Application (Required)

- A working demo — web app or mobile — covering all four pillars
- Must be runnable from source by the judging panel

4.2 Code Repository (Required)

- Public GitHub repository with all source code committed before the deadline
- README.md with: setup steps, required environment variables, how to run locally
- Architecture diagram showing data flow from CV upload through to agent response

4.3 Demo (Required)

- 5-minute recorded video
- **Must show this full flow:** CV upload → job search → fit score → AI assistant query → cover letter draft → tracker update

5. Bonus Points

Bonus Feature	What We Check
Live Deployment — App is accessible at a public URL (Vercel, Railway, Render, etc.). Must be stable and functional during judging.	Live URL reachable; core features work on hosted version
System Design Document — Written or diagrammed doc covering: data flow, how this scales to 10,000 users, estimated cost per user/month, key bottlenecks identified.	Depth and accuracy of the analysis
Evaluation Suite — At least 5 documented test cases: input, expected output, actual output, and pass/fail verdict.	Completeness and correctness of test cases

6. Tech Stack — You Choose

You have full freedom to pick your own stack. Below is a reference of what others have used successfully for each layer.

7. Rules

Permitted

- Open-source libraries, frameworks, and public UI component libraries
- Public APIs and third-party LLM providers
- Boilerplate starters (Next.js, CRA, etc.) — the core functionality must be built during the hackathon

Not Permitted

- Projects started before the hackathon kickoff
- Purchased pre-built software or submitting someone else's work

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- Hardcoded AI responses or faked live agent functionality
 - Sharing solutions with other teams

Try to do your project as closely as possible. You don't need to be perfect. If you can extend it further, you can. Any kind of creativity will be valued properly.

Build something you'd actually use.

The best submission won't just impress judges — it will be a tool that genuinely changes how a job seeker navigates their career. Good luck.